Dream Home AI

Design Choices & Methodology

Snaphomz Hackathon 2.0 — Project #1

June 6, 2026

Project Overview

Dream Home AI is an AI-powered web application that helps users discover their ideal home through a personality-based approach. Instead of traditional property search (filter by price, beds, location), users take a 2-minute quiz that maps their **lifestyle, taste, and goals** to a unique **Home Personality Type**, then receives:

- A personalized personality card with traits and style preferences
- AI-generated room visualizations matching their taste
- Top 5 property matches from 50 listings, scored across 5 dimensions
- Shareable results (PNG download, Twitter, copy link)

Core Thesis: *"Discover the home that matches your lifestyle before you ever search for a property."*

System Architecture

The application follows a **three-layer architecture** deployed on Vercel:

Layer	Technology	Purpose
Presentation	Next.js 16, Tailwind CSS, Framer Motion	UI rendering & animations
API	Next.js API Routes (serverless)	Business logic endpoints
Data	Local JSON + scoring engines	Properties, personalities

Data Flow:

1. User completes 8-step quiz → answers stored in `sessionStorage`
2. `POST /api/quiz/analyze` → Personality Engine scores answers → returns personality type
3. `POST /api/generate-images` → Returns AI or fallback room visuals
4. `POST /api/recommendations` → Matching Engine scores 50 properties → returns top 5

Methodology

3.1 Personality Engine (Weighted Scoring)

Quiz answers are scored against 10 pre-defined personality types using a **weighted multi-dimensional scoring model**:

Dimension	What It Measures	Weight
Lifestyle	Urban, suburban, nature, social, creative	25%
Architecture	Modern, traditional, minimalist, industrial, mediterranean	25%
Weekend Activity	How user spends leisure time	20%
Work Style	Office, remote, hybrid, entrepreneur, freelancer	15%
Must-Have Features	Pool, garage, garden, home office, etc.	15%

Each personality type defines which values it matches on each dimension. The type with the **highest composite score** is assigned. This approach is **deterministic, explainable, and instant** — no API calls needed.

3.2 Property Matching Engine (Multi-Factor Scoring)

Each of the 50 properties is scored against the user's profile across 5 factors:

Factor	**Scoring Logic**	**Weight**
Style Match	Exact style = 100, related = 50, other = 10	25%
Budget Match	Within range = 100, $\pm$20% = 60, $\pm$40% = 30	25%
Lifestyle Match	Tag overlap with lifestyle/weekend/work	20%
Space Match	Sqft alignment with size preference	15%
Feature Match	Must-have amenity coverage	15%

The weighted total produces a **match percentage** (0–100%) per property. Users see the top 5 with a visual breakdown of *why* each property matched.

3.3 Demo Resilience (Fallback Strategy)

A critical design goal: **the demo must never break**. We achieve this through:

- **Personality Engine:** Runs locally — zero external dependencies
- **Matching Engine:** Uses local JSON data — always available
- **Image Generation:** Tries OpenAI DALL-E 3 → falls back to curated Unsplash images per room/style combination
- **Share Card:** Uses `html-to-image` (client-side) — no server needed

Key Design Choices

Decision	**What We Chose**	**Why**
Framework	Next.js 16 (App Router)	SSR for SEO, serverless API routes, Vercel-native
Styling	Tailwind CSS 4 + dark theme	Rapid prototyping, luxury aesthetic
UI Library	ShadCN (Radix-based)	Customizable, lives in our codebase
Animations	Framer Motion	Production-grade React animations
State Mgmt	React Context + sessionStorage	Simple linear flow, no global store needed
Data Layer	Local JSON (50 listings)	Zero infrastructure for hackathon
Auth	None	Frictionless — every auth step loses ~30% users
Database	None	Stateless computation, all client-side
Deployment	Vercel (zero-config)	Built Next.js — best deployment experience
Design Language	Glassmorphism, dark mode	Premium brand positioning, modern 2024–2026 trend

UI/UX Design Rationale

- **Dark Theme Default:** Communicates luxury and premium positioning, consistent with high-end real estate brands.

- **Glassmorphism:** Frosted-glass card effects create visual depth without heavy shadows.
- **Gradient Accents:** Blue-to-purple gradients signal innovation and AI technology.
- **Micro-Animations:** Hover effects, page transitions, and loading states make the app feel alive and responsive.
- **Auto-Advance Quiz:** Single-select questions advance automatically, reducing clicks and friction.
- **Staged Loading:** Results load in 3 visible stages (analyzing → designing → matching), building anticipation.
- **Shareable Personality Card:** Designed for social virality — each share is free marketing for Snaphomz.

Revenue Connection to Snaphomz

1. Every matched property card has a **"View on Snaphomz"** CTA button
2. Results page includes a prominent **"Browse Snaphomz Listings"** banner
3. The **Repository Pattern** allows swapping mock data for live Snaphomz API — zero frontend changes needed
4. Shareable personality cards drive organic traffic back to the platform

Technology Stack

Category	**Technology**
Framework	Next.js 16 (App Router, TypeScript Strict)
Styling	Tailwind CSS 4, ShadCN UI
Animations	Framer Motion (motion v12)
Image Generation	OpenAI DALL-E 3 + Unsplash fallback
Image Export	html-to-image (client-side PNG)
Validation	Zod
Deployment	Vercel (serverless, edge)

Future Scope

- Replace mock data with **Snaphomz Live API** via Repository Pattern
- Add user accounts and saved results (**NextAuth.js**)
- Quiz analytics dashboard (**Supabase**)
- AI image caching (**Cloudinary/S3**)
- A/B testing personality result variations
- Multi-language support (i18n)

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